

NATIONAL TELE-OPHTHALMOLOGY AI SCREENING PROGRAM

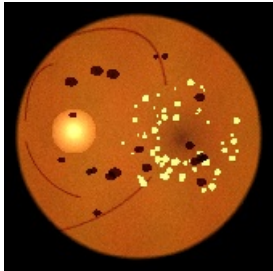
Explainable AI Clinical Decision Support System | Rural PHC Triage (SIH PS 26038)

Patient ID: IND-KA-PHC-8921 Age/Sex: 58 / Female Screening Date: 04-Sep-2026 23:08
PHC Health Centre: PHC Mulbagal, Kolar District Frontline Operator: ASHA Worker #108

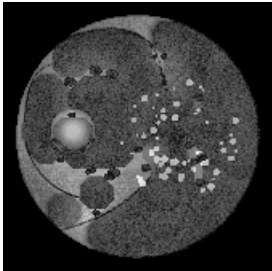
TRIAGE ACTION: URGENT SPECIALIST REFERRAL REQUIRED (<48 HOURS)

High risk of irreversible vision loss. Tertiary Vitreoretinal evaluation mandatory.

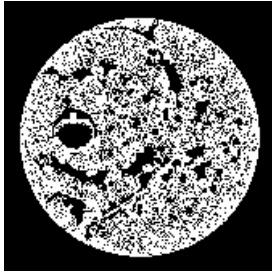
1. Raw Fundus Scan



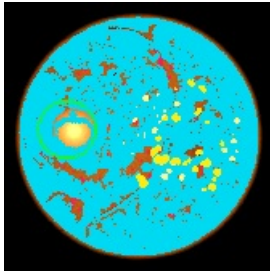
2. Enhanced Green



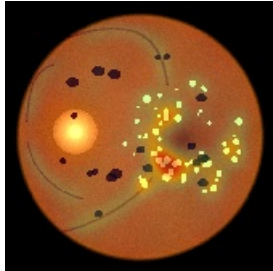
3. Retinal Vessels



4. Lesion Detections



5. Grad-CAM XAI

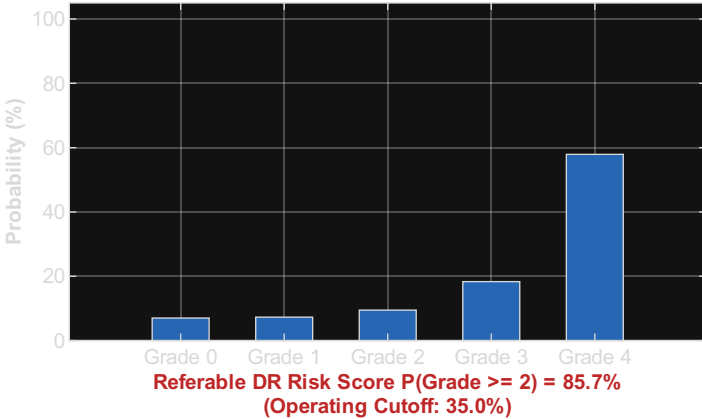


Legend: [Cyan]: Vascular Tree | [Yellow]: Hard Exudates (Lipids) | [Magenta]: Microaneurysms/Hemorrhages | [Jet Colormap]: AI Focus Regions

QUANTITATIVE BIOMARKER ANALYSIS

Image Quality Assessment (IQA): PASS (BRISQUE: 39.7)
Mean Illumination Level: 43.0% (Dynamic Range: 0.42)
Vascular Density (Coverage): 63.35% of FOV
Hard Exudates (Bright Lesions): 26 clusters (1688 px area)
Microaneurysms (Early NPDR): 54 detected foci
Retinal Blot Hemorrhages: 1 detected sites
Optic Disc Status: Localized (Center: [120, 243])

AI DR Severity: Proliferative DR (Conf: 57.9%)



OPHTHALMOLOGIST TELEMEDICINE REVIEW & VALIDATION (Time required: <30 seconds)

Clinical Impression: ☐ Concordant with AI diagnosis ☐ Upgrade Severity ☐ Downgrade / Artefact

Action Prescribed: ☐ Tele-Consultation Booked ☐ Anti-VEGF / Laser Referral ☐ PHC Annual Rescreen

Reviewing Ophthalmologist Signature: _____ Date/Time: _____